FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Advanced)-2020

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C

Sol. $H=\frac{u^2}{2g}.....(1)$

$\frac{u^2}{4}=u^2-2gh \Rightarrow h=\frac{3u^2}{8g}=\frac{3H}{4}$

2. C

Sol. $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}\ \ \frac{1}{-50}+\frac{1}{20}=\frac{1}{f}\ \ f=\frac{100}{3}$ $R = 2f = 66.67$ cm

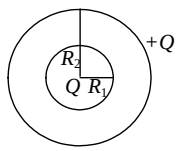
3. D

Sol. $\phi=\vec{B}.\vec{A}$ and $\varepsilon=-N\frac{d\phi}{dt}$ since angle between *B* and *A* is constant and magnitudes of *B* and A are also constant, so emf induced is zero.

4. D

Sol. Potential due to point charge and charge on the inner surface will cancel out each other, thus

$V=\frac{KQ}{R_2}$



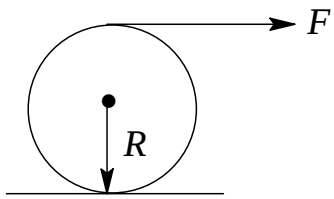
5. A

Sol. $PV = nRT \Rightarrow P = \left(\frac{nR}{V}\right)T \quad \therefore$ slope $\alpha\, n \;\therefore \quad n_A > n_B$

6. D

Sol. $\tau = F \times R = I\alpha$

In one full rotation, $\theta = 2\pi = \frac{1}{2}\alpha t^2 \Rightarrow t^2 = \frac{4\pi}{\alpha} = \frac{4\pi I}{FR}$



Linear acceleration, $a = F / M$

Distance travelled in one rotation = $2\pi R\left(given\right)$

$= \frac{1}{2}at^2 = \frac{1}{2}\frac{F}{M}.\frac{4\pi I}{FR} = \frac{2\pi I}{MR}$. It is a hollow cylinder.

7. A

Sol. Torque = $\tau = \frac{9}{10}mg\left(\frac{9}{20}L\right) = I\alpha = \frac{\left(\frac{9}{10}m\right)}{3}\left(\frac{9}{10}L\right)^2\alpha \Rightarrow \alpha = \frac{5}{3}\frac{g}{L}$

Acceleration, $a_{CM} = \frac{9L}{10}\alpha$

Now, $\frac{9}{10}mg - N_A = ma_{CM} \Rightarrow N_A = \frac{3}{20}mg$

8. AC

Sol. For translation equilibrium

$\frac{V}{4}\rho g = Mg - - - - - (1)$

$\frac{3V}{4}\rho g = Mg + mg - - - - - (2)$

$m = 2M$

For rotational equilibrium centre of mass of the system should lie below centre of buoyancy.

As $\frac{3\ell}{8}\left(\text{position of centre of buoyancy from lower end of rod}\right) > \ell/6$ (position of centre of mass form lower end of rod). So rotational equilibrium is possible.

9. AC

Sol. $\frac{dU}{dx} = (x+1)(1) + (x+2)1 = 2x + 3 = 0$

$x = -\frac{3}{2}$ is the equilibrium position

$\frac{d^2U}{dx^2} = 2 > 0$ so equilibrium stable

10. BD

Sol. $\frac{dK}{dx}=\frac{\mu V_p^2}{2}$ and $V_p=0$ at crest and trough

$\frac{dU}{dx}=\frac{I}{2}\left(\frac{\delta y}{\delta x}\right)^2$ and $\frac{\delta y}{dx}=0$ at crest and trough

11. AC

Sol. $\frac{KA(4-0)}{2-y}=\frac{3KA\{(0)-(-2)\}}{y}$

y=1.2m=120 cm

12. AC

13. D

Sol. $\tan\alpha=\frac{E_2}{E_1}=\frac{Kp\sin\theta/r^3}{2Kp\cos\theta/r^3}=\frac{1}{2}\tan\theta.....(1)$

$\alpha+\theta=\frac{\pi}{2}.....(2)$

Solving (1) and (2) $\tan\theta=\frac{1}{\sqrt{2}}$

14. ACD

Sol. $A\left(\frac{dB}{dt}\right)=\frac{L\,di}{dt}\Rightarrow\int_0^i di=\frac{A}{L}\int_0^B\frac{dB}{}\Rightarrow i=\frac{A}{L}B$

15. B

16. A

Sol. (for Q. 15 to 16)

$5mg\sin\theta-f=5ma.....(1)$

$fR=I\alpha.....(2)$

$a=R\alpha.....(3)$

$I=\frac{4m\ell^2}{3}+mR^2.....(4)$ and $\ell^2=3R^2$

Solving above equation

$a=\frac{15}{26}g\sin\theta$

and

$\mu mg\cos\theta=f=\frac{5mg\sin\theta}{\frac{1+5mR^2}{I}}$

$\mu=\frac{11}{26}\tan\theta$

17. B

18. C

Sol. (for Q. 17 to 18)

$$ma = F = \frac{-dU}{dx} = -32(x-1)$$

$$a = -16(x-1) \Rightarrow \omega = 4\,\text{rad/second}$$

At mean position $a = 0 \Rightarrow x = 1(\text{mean position})$

Applying conservation of mechanical energy

$$0 = -16 + K.\varepsilon_{\max} \Rightarrow K.\varepsilon_{\max} = 16\,\text{Joule}$$

Chemistry

PART – II

SECTION – A

19. A

Sol. $n=\frac{PV}{RT}=\frac{6\times2}{0.08\times300}=0.5mol$

$q=0$

$\Delta U = q + W$

$n\times3R\times(T_2-T_1)=-P_{ext}(8-2)$

$0.5\times3\times0.08(T_2-300)=-1atm\times6L$

$(T_2-300)=-50$

$T_2=250$

Now,

$2L,300\,K\longrightarrow8L,300K\longrightarrow8L,250K$

$\Delta s = nR\ln\frac{8}{2}$ $\quad$ $\Delta s = nC_V\ln\frac{250}{300}$

$=0.5\times8.0\times\ln4$ $\quad$ $=0.5\times3R\ln\frac{5}{6}$

$=8\ln2$ $\quad$ $=12\ln\frac{5}{6}$

$=8\times2.3\times0.3$ $\quad$ $=12[\ln10-\ln3-\ln4]$

$=12\times2.3[\log10-\log3-\log4]$

$=2.4\times2.3$ $\quad$ $=12\times2.3\times[-0.08]$

$=5.52$ $\quad$ $=-2.3\times0.96$

$=-2.208$ $\quad$ $\Delta S_{Total}=-3.3$

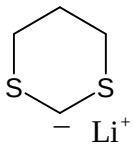
20. A

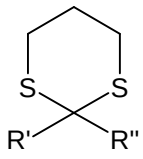
Sol. $BaO_2.8H_2O+H_2SO_4\longrightarrow BaSO_4\downarrow+H_2O_2+8H_2O$

$H_2O_2+H_2SO_4+2KI\longrightarrow K_2SO_4+2H_2O+I_2$

21. D

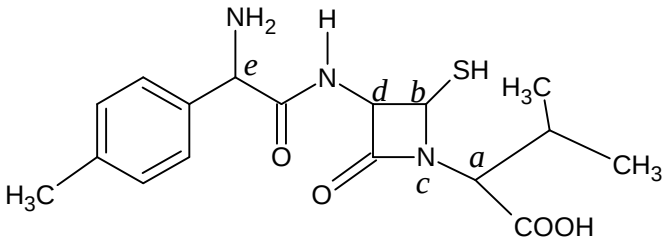
Sol.

P 

T= 

22. A

Sol. a, b, c, d, e are stereogenic centre.



23. A

Sol. $\frac{2.3\times10^{-15}}{284}\times N_A \text{ cm number of molecules}$

Cross-section of acid molecule $= \frac{100\times284}{2.3\times10^{-5}\times N_A}$

$= 2.05\times10^{-15}$

24. C

Sol. In equilibrium, vapour pressure of solution will remain same

$$\frac{\Delta p}{p^*} = \left(\frac{n_{sugar}}{n_{sugar}+n_{H_2O}}\right)_{\text{Beaker B}} = \left[\frac{2\ n_{NaCl}}{2\ n_{NaCl}+n_{H_2O}}\right]_{\text{BeakerA}}$$

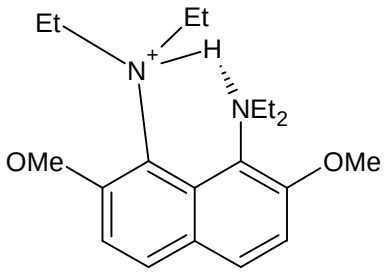
$$\frac{0.1}{0.1+\frac{(1-a)\times1000}{18}} = \frac{2\times0.1}{2\times0.1+\frac{(1+a)\times1000}{18}}$$

$\Rightarrow a = 1/3$

So molarity of NaCl $= \frac{0.1}{\frac{4}{3}} = \frac{0.3}{4} = 0.075$

25. A

Sol.



H-bonding

26. ACD

Sol. (A) As reaction is endothermic therefore it will go in the forward direction hence moles of CaO will increase.

(C) With the increase or decrease of volume particle pressure of the gases will remain same.

(D) Due to the addition of inset gas at constant pressure reaction will proceeded in the direct in which more number of gaseous are formed.

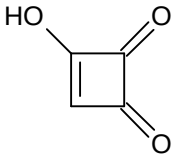
27. ACD

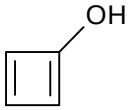
Sol. $W_{\text{on the system, adiabatic}} > W_{\text{on the system, isothermally}}$

$MgCO_3(s) \longrightarrow MgO(s) + CO_2(g)$

28. ABD

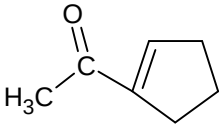
Sol. $A \rightarrow$ hydrogen of diketone is more acidic.

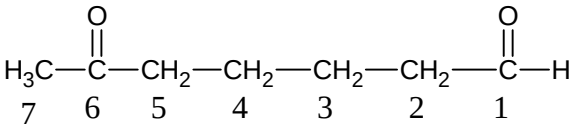
$B \rightarrow$  has some aromatic character

$C \rightarrow$  is antiaromatic

29. AB

Sol. (A)

 Aldol



Here C-2 and C-5 can form carbanion & leads to different product.

30. BCD

Sol. Aniline can show Carbylamine test, Diazo test, while Benzamide shows reaction with $NaOH+Br_2$.

31. BCD

Sol. Lyophilic colloids are used to protect lyophobic colloid.

32. BD

Sol. A.O. is a single e^- wave function

Area of the plot for r<52.9 pm is smaller than the area of the plot of r>52.9pm.

At higher atomic number every of $3d<4s$

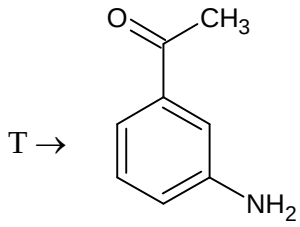
$$\Delta E_{2\to5} = RhcZ^2\left[\frac{1}{2^2}-\frac{1}{5^2}\right] = RhcZ^2\left[\frac{21}{100}\right]$$

$$\Delta E_{2\to4} = RhcZ^2\left[\frac{1}{2^2}-\frac{1}{4^2}\right] = RhcZ^2\times\frac{3}{16}$$

$$\Delta E_{2\to5} = \frac{28}{25}\Delta E_{2\to4}$$

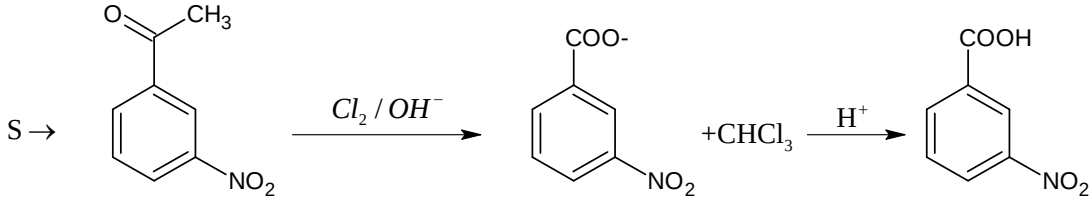
33. C

Sol.



34. B

Sol.



35. C

Sol. $SC^{3+} \rightarrow 3d^{\circ}$ so no scope of $M \rightarrow L$ bonding

36. D

Sol. $\left[Mn(CO)_6\right]^{+} \Rightarrow 25-1+12=36=$ EAN Rule

Mathematics

PART – III

SECTION – A

37. C

Sol. Let $z = r(\cos\theta + i\sin\theta)$

Then the equation $r^2 + \frac{4}{r^2} - 2.2\cos 2\theta - 16 = 0$

$r^4 - 4r^2(\cos 2\theta + 4) + 4 = 0$

By solving we get

For maximum value of r, $\cos 2\theta = 1$

$r = 2 + \sqrt{6}$

38. A

Sol. Equation of PX$\rightarrow(1+t^2)y = 2t(x+a)$ …(1)

For point Q, m solving equation (1) & curve

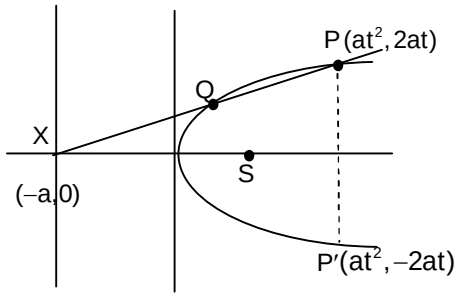
$t^2(x+a)^2 = ax(1+t^2)^2$

$(x - at^2)(xt^2 - a) = 0$

$x = \frac{a}{t^2} \Rightarrow \therefore Q = \left(\frac{a}{t^2}, \frac{2a}{t}\right)$

Equation of $P'Q \Rightarrow (x-a) + \left(\frac{t^2-1}{2t}\right)(y) = 0$

Will always passes through (a,0)



39. B

Sol. $g(x) = \frac{2(\sin x - \sin^n x) + |\sin x - \sin^n x|}{2(\sin x - \sin^n x) - |\sin x - \sin^n x|}$

for $0 < n < 1$, $\sin x < \sin^n x$, so $g(x) = \frac{1}{3}$ and for $n > 1$, $\sin x > \sin^n x$, so $g(x) = 3$

$\therefore$ for $n > 1$, $f(x) = 3$, $g(x) = 3$

$\therefore$ f(x) is continuous and differentiable at $x = \frac{\pi}{2}$ and for $0 < n < 1$

$$f(x) = \begin{cases} \left[\frac{1}{3}\right] = 0 & x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right) \\ 3, & = \frac{\pi}{2} \end{cases}$$

$\therefore$ f(x) is not continuous at $x = \frac{\pi}{2}$. Hence f(x) is also not differentiable at $x = \frac{\pi}{2}$.

40. D

Sol. Given relation can be written as

$\frac{1}{8}(2\cos^2 5x - \cos 2x + \cos 8x - \cos 4x + \cos 6x) < \frac{6}{8} < \frac{3}{4}$

Therefore, no solution

41. B

Sol: $2\cos\frac{\pi}{7}+9\cos\left(\frac{4\pi}{21}\right)=-(6\cos a+7\cos b)$

$2\sin\frac{\pi}{7}-9\sin\frac{4\pi}{21}=6\sin a-7\sin b$

Squaring and adding, we get

$42\cos(a+b)=9$

42. A

Sol. $2\cos10^\circ+\sin100^\circ+\sin1000^\circ+\sin10000^\circ$

$=2\cos10^\circ+\sin\left(90^\circ+10^\circ\right)+\sin\left(3\times360^\circ-80^\circ\right)+\sin\left(27\times360^\circ+280^\circ\right)$

$=2\cos10^\circ+\cos10^\circ-\cos10^\circ-\cos10^\circ$

$=\cos10^\circ$

43. ACD

Sol. Placing 6 different colour balls into 3 different size boxes is as the number of onto function form 6 elements set to 3 element sets.

$\therefore$ Total number of ways $3^6-3.2^5+3.1^4$ (By inclusion – Exclusion principle)

Or

$\left({}^6C_4+\frac{6!}{3!2!}+\frac{6!}{(2!)^3 3!}\right)3!$

44. ACD

Sol. $|A|=3abc-a^3-b^3-c^3$

$a+b+c=9.\&\,ab+bc+ca=26$

So, $|A|=-27$

Hence $M=3^{48}$

Now $3^{48}=9^{24}=(10-1)^{24}$

Now last two digits are 61.

45. ACD

46. BC

Sol. Since x = 1, 2, 3 are normals

$\Rightarrow$ f'(x) = $4ax^3 + 3bx^2 + 2cx + d$

$\Rightarrow$ b = – 8a, C = 22a, d = –24a

Also f(2) > f(0)

$\therefore$ x =1 & 3 are point of local maxima

Also f(k) = f(0)

$\Rightarrow ak^4 + bk^3 + ck^2 + dk = 0$

$\Rightarrow ak\,(k-4)\,(k^2-4k+6)=0$

$\Rightarrow k = 4$

47. ABCD

Sol. $I_n = \int_0^{\pi/4} \tan^n x\, dx$

$= \int_0^{\pi/4} \tan^{n-2} x\left(\sec^2 x - 1\right)dx$

$= \frac{1}{(n-1)} - I_{n-2}$

Or, $I_n + I_{n-2} = \frac{1}{(n-1)}$

48. D

Sol: $\int \frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}}\,dx = \int \frac{x^2-1}{x^5\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}}\,dx = \int \frac{\frac{1}{x^3}-\frac{1}{x^5}}{\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}}dx$

Let $2-\frac{2}{x^2}+\frac{1}{x^4} = t^2$

49. AD

Sol. $\therefore P(X=4), P(X=5)$ and $P(X=6)$ are in AP.

$\therefore 2P(X=5) = P(X=4) + P(X=6)$

$\Rightarrow \quad 2 = \frac{P(X=4)}{P(X=5)} + \frac{P(X=6)}{P(X=5)}$

$2 = \frac{{}^nC_4}{{}^nC_5} + \frac{{}^nC_6}{{}^nC_5}$

$\Rightarrow n^2 - 21n + 98 = 0 \quad \therefore n = 7, 14$

50. B

Sol. Let R be the event that a red face appears in each of the first n throws,

E_1: Die A is used when head has already fallen

E_2: Die B is used when tail has already fallen

$\therefore P\left(\frac{R}{E_1}\right) = \left(\frac{2}{3}\right)^n \& P\left(\frac{R}{E^2}\right) = \left(\frac{1}{3}\right)^n$

As per the given condition,

$\frac{P(E_1).P\left(\frac{R}{E^2}\right)}{P(E_1).(PR/E^1) + P(E_2).P\left(\frac{R}{E_2}\right)} = \frac{32}{33}$

$\frac{\frac{1}{2}.\left(\frac{2}{3}\right)^n}{\frac{1}{2}\left(\frac{2}{3}\right)^n + \frac{1}{2}.\left(\frac{1}{3}\right)^n} = \frac{32}{33}$

$n = 5$

51. C

Sol. $\det(A_0)\det\left(Al + A_0B_0^2 + A_0^2 + A_0^3B_0^4 + ...\text{up to 10 terms}\right) = 0\left(\because \det(A_0) = 0\right)$

52. C

Sol. $B_1 = adj\left(B_0\right) = B_0$

Hence $B_1 = B_2 = B_3 = ...B_{494} = B_0$

$\therefore B_1 + B_2 + ... + B_{49} = 49B_0$

53. A

Sol. Required probability $= \frac{1}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{2}\left(1 - \frac{1}{4}\right) = 0.728$

54. A

Sol. $P\left(\frac{\text{I}}{\text{II}}\right) = \frac{P(\text{I}).P\left(\frac{\text{II}}{\text{I}}\right)}{P(\text{II})} = \frac{\frac{1}{2} \cdot \frac{1}{\sqrt{2}}}{\frac{1}{2} \cdot \frac{1}{\sqrt{2} + \frac{1}{8}}} = 0.74$